Practice workbook · Identify the independent or dependent variable

A described method or results table where the student names the factor deliberately changed, the quantity measured, or both.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-IDENTIFY-INDEPENDENT-DEPENDENT-VARIABLE

1. Question 4 · part b.ii · 2 marks

June 2022 Paper 1R

Worked-example retry · Attempt part b.ii only.

4 (a) Lipids are digested in the small intestine.

(i) Describe a test for lipids in food.

(3)

(ii) Name the enzyme that digests lipids.

(1)

(iii) Name the products of lipid digestion.

(2)

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(b) A student investigates how the pH of a lipid solution, containing an enzyme that digests lipids, changes over 10 minutes at two different temperatures.

The tables show the student's results.

Temperature at 37 °C

Time in minutes	1	2	3	4	5	6	7	8	9	10
pH	8.5	8.0	7.5	7.2	7.0	6.8	6.2	5.8	5.5	5.5

Temperature at 80 °C

Time in minutes	1	2	3	4	5	6	7	8	9	10
pH	9.2	9.0	8.8	8.8		8.6	8.6	8.4	8.4	8.4

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1. Question 4 · part b.ii · 2 marks

June 2022 Paper 1R

Worked-example retry · Attempt part b.ii only.

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(i) Estimate a pH value for the lipid solution at 80 °C at 5 minutes.

(1)

(ii) Give the independent variable and the dependent variable for this investigation.

(2)

independent

dependent

(iii) Explain the differences in the results of the student's investigation at the two different temperatures.

(4)

(Total for Question 4 = 13 marks)

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1. Question 4 · part b.ii · 2 marks · Working and answer

Working and answer

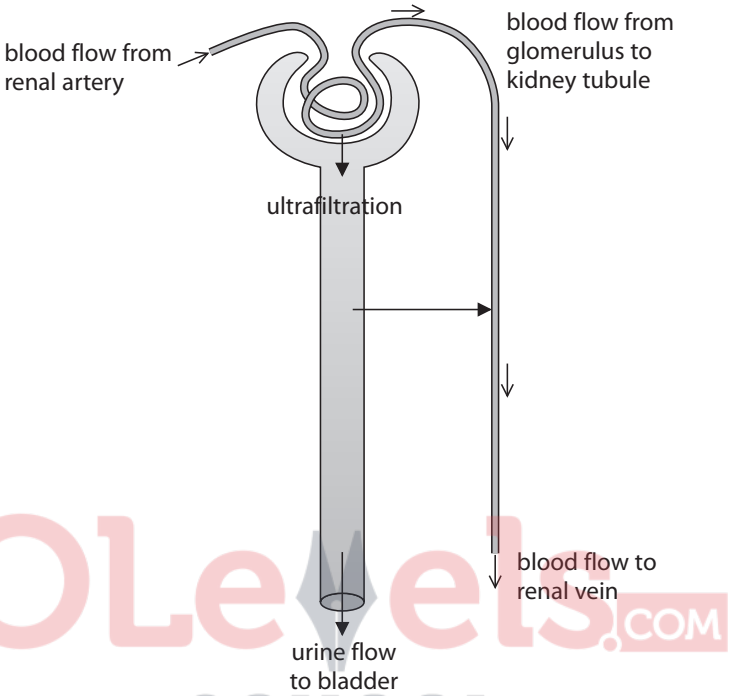
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2. Question 8 · part b.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.i only.

8 (a) The diagram shows part of a nephron in the human excretory system.



(Source: adapted from © Alila Medical Images/Alamy)

(i) Explain how ultrafiltration occurs in the Bowman's capsule.

(2)

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2. Question 8 · part b.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.i only.

(ii) Explain the differences in the composition of the blood flowing through the renal artery and through the renal vein.

(4)

(b) The quantity of salt in the diet affects the concentration of urine produced.

An investigation is set up to test this statement.

This is the method used.

- ten people are each given a different food
- each food contains a different quantity of salt
- the volume of urine produced by each person is recorded each hour for six hours after eating the food

(i) State the independent variable in this investigation.

(1)

(ii) Explain why this investigation is unlikely to produce valid results.

(2)

(iii) State one change to the method that would make the results more valid.

(1)

(Total for Question 8 = 10 marks)

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2. Question 8 · part b.i · 1 mark · Working and answer

Working and answer

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3. Question 4 · part a.iii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.iii only.

4 Aspirin is the active ingredient used in some tablets to relieve pain.

Aspirin becomes acidic when it dissolves in the body. The faster aspirin dissolves, the faster it can relieve pain.

A student investigates if it is better to mix sucrose or starch with the active ingredient to make an aspirin tablet.

This is the student's method.

- add distilled water to three beakers, A, B and C
- add 1 aspirin tablet to each beaker
- add 5 g of sucrose to beaker B
- add 5 g of starch to beaker C
- measure the pH of the contents of each beaker at the start of the investigation
- measure the pH of the contents of each beaker every minute for 5 minutes

The table shows the student's results.

Beaker	pH of contents of beaker					
	at start	1 minute	2 minutes	3 minutes	4 minutes	5 minutes
A – aspirin	7.6	7.1	6.6	6.2	5.8	5.4
B – aspirin and sucrose	7.2	6.7	6.3	5.7	5.2	4.7
C – aspirin and starch	7.1	7.2	7.3	7.3	7.4	7.4

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3. Question 4 · part a.iii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.iii only.

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(a) (i) Plot a graph of the student's results.

Join the points with straight lines.

(5)



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(ii) State two variables that should be controlled in this investigation.

(2)

1 .....

2 .....

(iii) Name the dependent variable in this investigation.

(1)

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3. Question 4 · part a.iii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.iii only.

(b) Explain how the results of this investigation could help a manufacturer improve the effectiveness of its aspirin tablets.

(3)

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(Total for Question 4 = 11 marks)

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Working and answer

4. Question 7 · part a.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.ii only.

7 Garlic plants contain a substance called allicin. This substance is thought to have antibacterial properties.

A student investigates the antibacterial effect of garlic.

This is the student's method.

- label three test tubes A, B and C
- add 10 cm^3 of milk into each of the tubes
- add a sample of the bacterium *E. coli* into tubes B and C using aseptic techniques
- crush the garlic in water to form garlic juice
- add 1 cm^3 of garlic juice to tube C
- shake all three tubes and leave to incubate for two hours
- place a sample from tube A into the centre of Petri dish A, from tube B into the centre of Petri dish B and from tube C into the centre of Petri dish C
- incubate the three Petri dishes for five days
- each day measure the diameter of any bacterial growth that has occurred

The table shows the student's results.

Petri dish	Diameter of bacterial growth in mm				
	day 1	day 2	day 3	day 4	day 5
A	0	1.7	3.0	4.6	7.1
B	0	4.2	8.4	15.1	36.5
C	0	0	0	0	0

(a) (i) State three control variables in the student's investigation.

(3)

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4. Question 7 · part a.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.ii only.

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(ii) State the dependent variable in the student's investigation.

(1)

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(b) (i) Calculate the mean daily increase in the diameter of the bacterial growth for dish A and dish B.

(2)

mean daily increase for dish A ..... mm/day

mean daily increase for dish B ..... mm/day

(ii) Explain the difference between the mean daily increases in the diameter of the bacterial growth for dish A and dish B.

(4)

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(c) Explain what is meant by an aseptic techniques for the transfer of bacteria.

(2)

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4. Question 7 · part a.ii · 1 mark · Working and answer

Working and answer

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5. Question 6 · part a · 1 mark

June 2022 Paper 2

Further practice · Attempt part a only.

6 A student uses this method to investigate the effect of different antibacterial cleaners on the growth of bacteria.

- add bacterial culture to sterile agar in a Petri dish using an inoculating loop
- using a pipette, place two drops of an antibacterial cleaner onto a filter paper disc
- repeat using two other filter paper discs, each one soaked with a different antibacterial cleaner
- soak one filter paper disc in distilled water
- place the four filter paper discs onto the sterile agar
- place a lid on the Petri dish
- place the dish into an incubator at 25 °C for 5 days

(a) State the independent variable in this investigation.

(1)

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(b) Give two improvements to this method that will reduce the risk of introducing unwanted microorganisms into the Petri dish.

(2)

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5. Question 6 · part a · 1 mark · Working and answer

Working and answer

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6. Question 5 · part a.ii · 1 mark

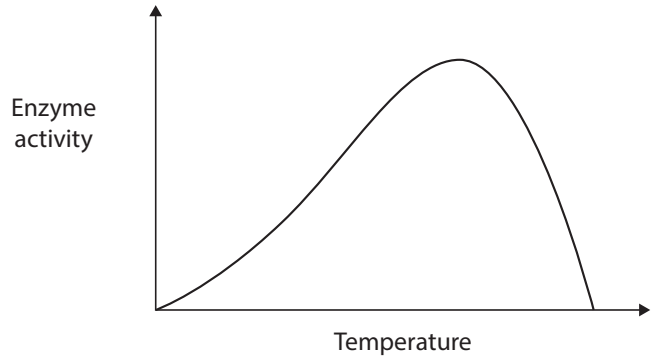
June 2024 Paper 1

Further practice · Attempt part a.ii only.

5 (a) A group of students investigate the effect of temperature on enzyme activity.

The students use the results they obtain to plot a graph.

The graph shows the students' results.



(Source: adapted from https://ib.bioninja.com.au/standard-level/topic-2-molecular-biology/25-enzymes/enzyme-activity.html)

(i) Design a method the students could use to obtain the results shown in the graph.

(5)

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6. Question 5 · part a.ii · 1 mark

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Further practice · Attempt part a.ii only.

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(ii) State the independent variable in this investigation.

(1)

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(b) Explain how enzyme activity is affected by low temperatures.

(4)

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(Total for Question 5 = 10 marks)

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Working and answer

7. Question 2 · part c · 1 mark

November 2024 Paper 1

Further practice · Attempt part c only.

2 A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

Time in minutes	pH of solution
1	6.8
2	6.4
3	6.0
4	5.8
5	5.6
6	5.2
7	5.0
8	4.8
9	4.6
10	4.6

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7. Question 2 · part c · 1 mark

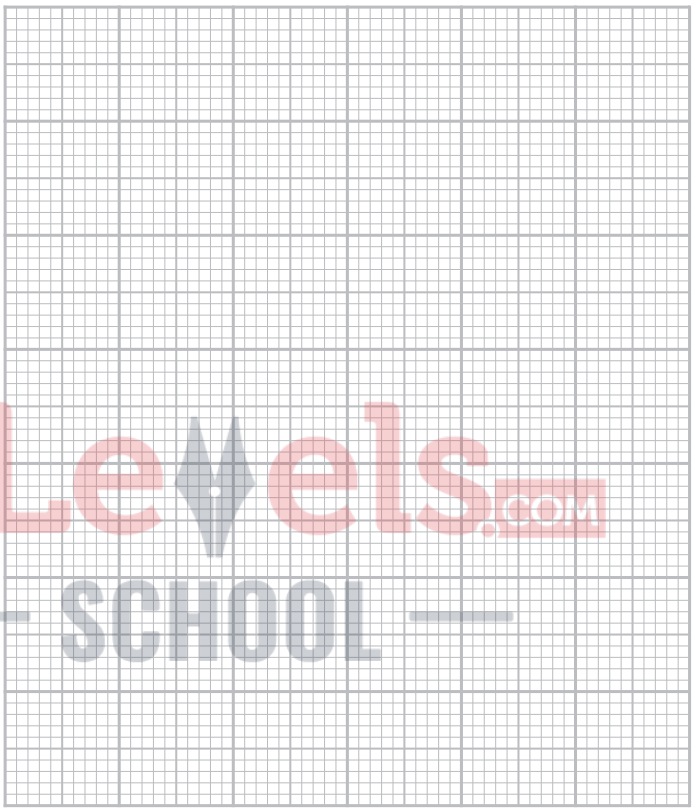
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Further practice · Attempt part c only.

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(a) (i) Plot a graph of the student's results.

(3)



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(ii) Draw a line of best fit.

(1)

(iii) Give the trend shown by the graph.

(1)

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7. Question 2 · part c · 1 mark

November 2024 Paper 1

Further practice · Attempt part c only.

(b) Explain the difference in the pH at 1 minute and the pH at 10 minutes.

(3)

(c) Name the independent variable in this investigation.

(1)

(d) Give two variables that the student should control in this investigation.

(2)

1 .....

2 .....

(e) State how the student could improve the reliability of their results.

(1)

(Total for Question 2 = 12 marks)

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7. Question 2 · part c · 1 mark · Working and answer

Working and answer

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8. Question 3 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

3 (a) The nervous system is made up of sensory, motor and relay neurones.

(i) Draw a diagram of a motor neurone.

You should label these structures in your diagram.

- nucleus
- axon
- myelin sheath

(4)

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(ii) Describe the function of a motor neurone.

(2)

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8. Question 3 · part b.ii · 1 mark

June 2025 Paper 2R

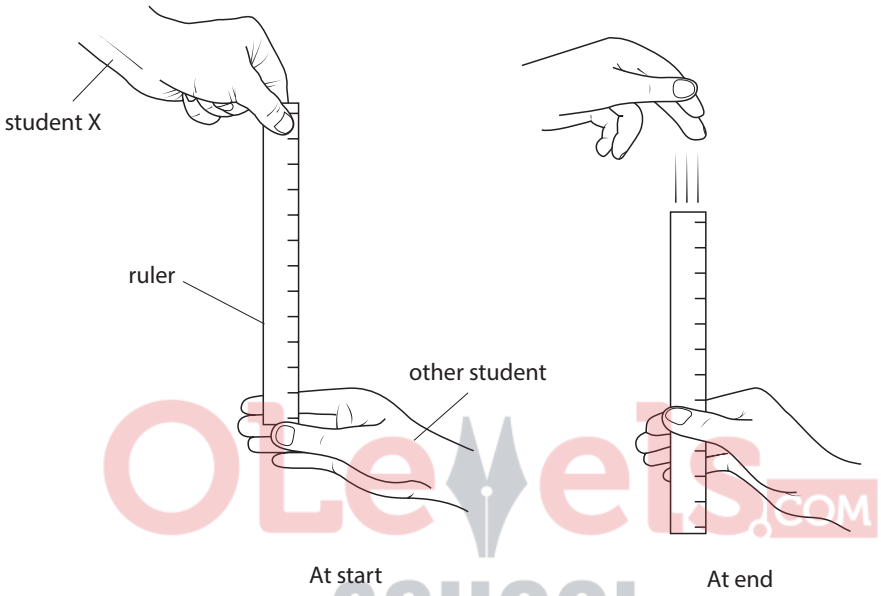
Further practice · Attempt part b.ii only.

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(b) Student X uses this equipment to investigate the reaction time of another student.



Student X knows that the reaction time can be calculated using this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

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8. Question 3 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

(i) Describe a method the student could use to compare the reaction times of 10 students.

(6)

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(ii) Name the dependent variable in this investigation.

(1)

(iii) Give a reason why the student should display the results using a bar chart instead of a line graph.

(1)

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8. Question 3 · part b.ii · 1 mark · Working and answer

Working and answer

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9. Question 7 · part b · 2 marks

November 2024 Paper 2

Further practice · Attempt part b only.

7 Six students investigate the activity of an enzyme found in the alimentary canal.

This enzyme digests a cloudy suspension into a clear solution.

The results of the investigation are shown in the table.

Student	Volume of enzyme solution in cm^3	Temperature in °C	pH	Volume of cloudy suspension in cm^3	Time tubes left in minutes	Appearance at end of investigation
1	5	37	2	10	9	cloudy
2	5	37	2	10	20	clear
3	5	37	4	10	35	clear
4	5	37	4	10	20	cloudy
5	5	37	6	10	8	cloudy
6	5	37	6	10	40	cloudy

(a) Explain why the investigation was carried out at 37 °C.

(2)

(b) Explain which property of the enzyme the students are investigating.

(2)

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9. Question 7 · part b · 2 marks

November 2024 Paper 2

Further practice · Attempt part b only.

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(c) Comment on the results of the investigation.

In your answer include ideas of validity and accuracy.

(4)

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(d) Describe a control experiment that the students could have carried out.

(3)

9. Question 7 · part b · 2 marks · Working and answer

Working and answer

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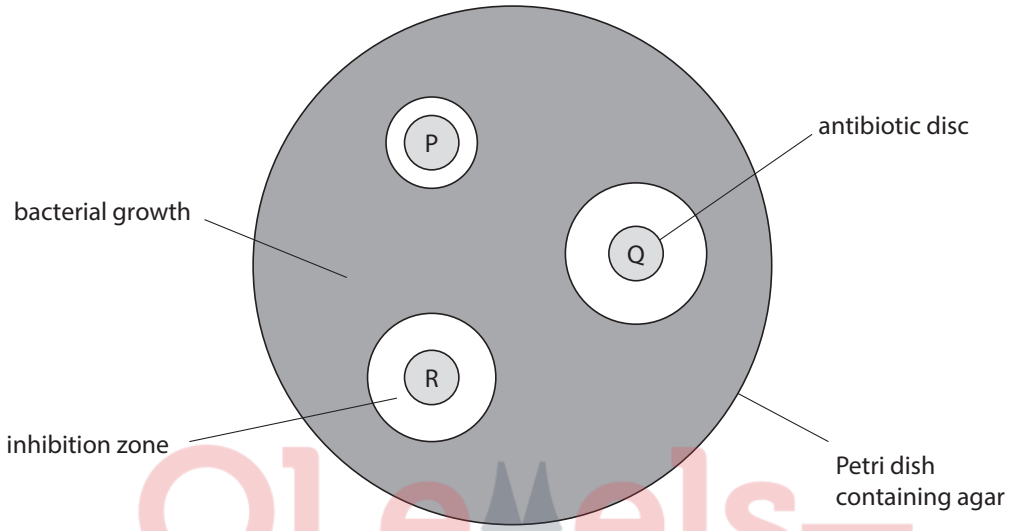
10. Question 5 · part a.i · 1 mark

June 2024 Paper 2

Further practice · Attempt part a.i only.

5 (a) A student investigates the action of three different antibiotics, P, Q and R, on the growth of bacteria.

The diagram shows the results of the investigation.



(i) Which is the independent variable in this investigation?

(1)

- ☒ **A** the mass of bacteria added
- ☒ **B** the size of the inhibition zone
- ☒ **C** the type of antibiotic added
- ☒ **D** the mass of agar in the Petri dish

(ii) Give two procedures that the student should use to reduce the risk of contamination by unwanted bacteria.

(2)

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2 .....

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10. Question 5 · part a.i · 1 mark · Working and answer

Working and answer

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